

M01 / LEARNER BOOK

Data and essential Excel

Standalone Zero-to-Job-Ready Data Engineer 2026

15 controlled lessons

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

START HERE | How to use this module 3

DE01-01 | What data is: tables, records, fields and data types 4

DE01-02 | Grain: what one row represents 11

DE01-03 | Keys, duplicates and missing values 14

DE01-04 | Data quality and useful validation 17

DE01-05 | Meet Excel: workbook, worksheet, cells and formulas 20

DE01-06 | Enter, save and structure tabular data safely 23

DE01-07 | Sort and filter without damaging records 26

DE01-08 | Basic formulas and essential functions 29

DE01-09 | Relative and absolute references 32

DE01-10 | Excel Tables and structured working habits 35

DE01-11 | Practical cleaning and exact lookups 38

DE01-12 | PivotTables and Power Query awareness 41

DE01-13 | Chart choice and basic visual communication 44

DE01-14 | Reconciliation, sanity checks and source preservation 47

DE01-15 | A complete small workflow: inspect, calculate, validate, explain and save 50

START HERE

How to use this module

Open the matching practice folder from the course root: 01_M00_M03/labs. Work on a copy. Commands printed in a lesson use that module as the working folder unless the lesson names a different location.

Study in lesson order. After the last lesson, use Quick Revision, take the Module Revision Test, then open its separate Protected Answers. Gates are additional assessments at the boundaries shown in the course guide.

This book is written for a beginner. Work through the lessons in order. Read the new words, watch the assigned instructor range, perform the printed example on your laptop, then save an independent attempt before opening the explained review. Ready-made revision is included; you do not need to rewrite the book as notes.

This cumulative package contains M00-M03, G01 and G02. It keeps the completed earlier modules alongside the new SQL modules. The full standalone course continues to M19; the later modules are not delivered as completed books in this batch. There is no fixed time-to-job promise.

Instructor videos open online and use their creators' data. Exact start and stop times are printed with each assigned range. Stop manually at the end: a YouTube start link does not enforce the stop. Published chapter titles, boundaries and runtime were inspected, not complete playback. Read the purpose and any software difference beside the link. A lesson can include written details that its supporting video does not cover; coverage limits remain explicit in WATCH_GUIDE.html.

Study without a Mentor app for now. Keep your own attempts and result evidence using the supplied template. The books contain learner tasks and changed retries; the separate review books contain their answers. The cumulative ZIP includes all required fictional inputs. Never open an answer script before saving your genuine attempt.

Practice workbook: labs/M01/M01_Practice.xlsx. Explained answer workbook: labs/M01/M01_Explained_Answers.xlsx. Their matching explained reviews are in M00_M01_Explained_Reviews.pdf. Attempt G01 after M01.

DE01-01

What data is: tables, records, fields and data types

Watch first

[1. What is Data? An Introduction to Data Terminology and Concepts](#)

00:00-06:38

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE00-08. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: What is Data? An Introduction to Data Terminology and Concepts, 00:00-06:38](#). Visual introduction to data, variables, datasets and databases. Relate each term to our order records; our book supplies storage types, leading-zero IDs and field-level examples. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

Tables, records, fields and data types

Why learn this before Excel, SQL or Python?

Imagine a stationery shop. The owner asks, "How much money is represented by these orders, and which orders are marked paid?" Before using a formula, you must understand what the saved information means. Otherwise, a perfectly typed formula can answer the wrong question.

Data is recorded information. "Someone bought notebooks" is information. Saving the order number, item, quantity, price, date and payment status makes that information easier to find, compare and check.

You will learn to read a small table, identify records and fields, choose sensible types, and explain why a missing value must not be guessed. You will also check a few simple amounts. No Excel, SQL or Python knowledge is needed for this book-based exercise. A calculator is fine.

Picture a stack of order cards

One card describes one order. Every card has the same labelled spaces. Put the cards into a grid: each card becomes one horizontal row; each labelled space becomes one vertical column.

Word	Plain-English meaning
Table	Information arranged into named columns and rows
Record / row	One complete entry about the thing we are recording
Field / column	One named kind of information collected for each record

Word	Plain-English meaning
Value	The particular information in one place, such as quantity 2
Cell	The place where one row and one column meet
Header	A column's label, such as quantity; it is not a customer order

Different tools sometimes use "field" more broadly. Here it means the named column. All inputs, worked reasoning, practice, answers and revision notes are supplied. You do not need to find a dataset or write your own lesson notes.

For this fictional example, one record means one order, and each order contains one product kind. Real orders can contain several product kinds; we will learn how to model that later. For now, keep the meaning of one row consistent.

order_id	item	quantity	unit_price	ordered_on	is_paid
0007	Notebook	2	49.50	2026-09-01	TRUE
0008	Pen	3	10.00	2026-09-01	FALSE
0009	Folder	1	25.00	2026-09-02	TRUE
0010	Marker	2	30.00	2026-09-02	(missing)

The printed marker (missing) means no payment status was supplied. It is not the source value FALSE or zero. Different tools represent missing information differently; the important point is that we do not yet know the answer.

Follow the first record with me

Start at order_id 0007 and move across the same row. The item is Notebook. The quantity is 2. The unit price is 49.50: the price of one notebook. The date is 2026-09-01. TRUE under is_paid means the order is recorded as paid.

Read it as a sentence: "Order 0007 contains two notebooks at 49.50 each, was ordered on 1 September 2026, and is marked paid." The values have meaning because you know which field each belongs to.

Four records, six fields

Count the order entries: 0007, 0008, 0009 and 0010. There are 4 records. The header is not a fifth order. Count the six labels across the top: there are 6 fields.

At the meeting point of order 0008 and quantity, the value is 3. At order 0008 and unit_price, the value is 10.00. Both are numbers, but quantity answers "how many?" and unit price answers "how much for one?"

Connect this to a spreadsheet

If this exact table starts at A1, its headers occupy A1:F1. Order 0007 is on spreadsheet row 2. Quantity is column C, so C3 contains 3 for order 0008. A spreadsheet row number is a position; order_id is a saved business label. They are different things.

Pause and say it: "A row tells me about one order. A column tells me one kind of fact about the orders."

A data type is a category of value. It helps us decide which values make sense and which operations are appropriate. Adding quantities can be useful. Adding order labels does not tell us anything useful about the orders.

Text stores characters, including letters and digit characters. An integer is a whole number. A decimal number can contain a fractional part, such as 49.50. A date represents a calendar date. A Boolean has two possible known values: true and false. Missing information is the absence of a known answer, not a third Boolean answer.

Field	Intended type	Rule and reason
order_id	Text	0007 is a label. Keep the leading zeros. Do not treat it as a quantity
item	Text	Notebook is a name, not a number
quantity	Integer	These items are counted in whole units; quantity must be positive here
unit_price	Decimal amount	49.50 includes part of a currency unit; preserve the intended precision
ordered_on	Date	2026-09-01 means year 2026, month 09, day 01
is_paid	Boolean; missing allowed	TRUE = recorded paid; FALSE = recorded unpaid; missing = no supplied status

Why 0007 should not become 7

If the printed card says 0007 and software changes it to 7, the values may look numerically equivalent, but the original label has changed. Labels can also contain letters, such as A007. We use text because identity, not arithmetic, is the field's job.

Type and appearance are different

49.5 and 49.50 represent the same amount in this example. Displaying two decimal places makes prices consistent; appearance alone does not establish the stored type. A cell displaying 2026-09-01 might still contain text until software interprets it as a date.

For now, choose the intended meaning. The full software companion must show how to inspect what the tool actually stored. Excel, PostgreSQL and Python do not use identical internal types.

Small check: Does "two" satisfy this example's integer quantity rule? No. A person understands the word, but it does not follow the agreed numeric form. A cleaning step might convert it; that

conversion must be explicit and checked.

The owner wants the value of all four orders. All prices use the same currency. Ignore taxes, discounts and delivery charges: they are not part of this example's rules.

Step 1: calculate what one row costs

Order amount = quantity x unit_price. Unit price tells us the price of one item; quantity tells us how many of that item were ordered.

For order 0007, calculate $2 \times 49.50 = 99.00$. Using 49.50 as the whole order amount would count only one of the two notebooks.

Order	Calculation	Amount	Payment category
0007	2×49.50	99.00	Known paid
0008	3×10.00	30.00	Recorded unpaid
Order	Calculation	Amount	Payment category
0009	1×25.00	25.00	Known paid
0010	2×30.00	60.00	Unknown status

Step 2: add the four order amounts

$99.00 + 30.00 = 129.00$. Add 25.00 to reach 154.00. Add 60.00 to reach 214.00. That is the value of all four orders under the stated rules.

Step 3: answer without guessing

Known-paid orders are 0007 and 0009: $99.00 + 25.00 = 124.00$. Recorded-unpaid order 0008 contributes 30.00. Order 0010 has no supplied payment status, so 60.00 stays in an unknown-status category.

We cannot say that every order not marked TRUE is unpaid. That turns "we do not know" into "we know the answer is no". Here, the difference affects 60.00 of order value.

Step 4: reconcile the categories

Reconcile means compare results that should agree and investigate any difference. Counts: 2 paid + 1 unpaid + 1 unknown = 4 records. Amounts: $124.00 + 30.00 + 60.00 = 214.00$, matching the whole-table total.

These checks show that our calculations and categories agree with the supplied table. They do not prove that every source value was correct in real life. A result can be consistent with its source while the source itself contains a mistake.

If you get this result...	Likely cause	How to repair it
"There are 5 orders"	Counted the header	Point to each order_id: only four rows describe orders

If you get this result...	Likely cause	How to repair it
Order 0007 becomes 7	Treated a label as numeric	Preserve the original text label; inspect the actual stored type in the software lab
Total is 114.50	Added unit prices	Calculate quantity x price for each row, then add the row amounts
Unpaid amount is 90.00	Treated missing as FALSE	Separate 60.00 unknown from 30.00 recorded unpaid
Paid amount is 214.00	Included every order	Include only rows explicitly marked TRUE

A correct total can hide a mistake

Suppose someone reports paid 124.00, unpaid 90.00 and unknown 0.00. These still add to 214.00. The total check passes, but the classification is wrong.

Inspect order 0010. Its payment status is missing, so it cannot support an "unpaid" conclusion. Check both the total and the category rule. A check should test the mistake you are worried about.

Use this recovery ladder if you feel lost

1. Meaning first: say what one row describes. Here, one order containing one product kind.
2. Find the field: do you need a quantity, a price, a label, a date or a status?
3. Use one record: read order 0007 aloud and calculate only that amount.
4. Compare: if your amount is not 99.00, fix that before adding every row.
5. Try another record: order 0008 has 3 items at 10.00 each. Explain why its amount is not just 10.00.

Why the checks are useful

A record-count check can catch a missing row. A total can catch some arithmetic errors. A category check can catch payment-status mistakes. No single check answers every question.

You are beginning to think like a data engineer: preserve what the source says, state the meaning of the data, and choose checks that can detect specific mistakes. This does not require complicated software. It starts with reading a small table carefully.

Try it yourself

Keep Review closed until you have made an attempt. A calculator is allowed. Say your answers aloud or type them into the supplied answer form. The inputs are different from the worked example.

Same rules: one row is one order containing one product kind; unit_price is the price of one item; prices share one currency. Ignore taxes, discounts and delivery charges. Missing status is not FALSE.

order_id	item	quantity	unit_price	ordered_on	is_paid
0011	Notebook	1	60.00	2026-09-03	TRUE
0012	Pen	4	12.50	2026-09-03	FALSE
0013	Ruler	2	20.00	2026-09-04	(missing)
0014	Bag	1	250.00	2026-09-04	TRUE

Your tasks

1. State what one row means, the record count and the field count. Explain why the header is not a record.
2. Find the quantity for order 0012. Explain how it differs from that order's unit price.
3. Choose an intended type for each field. Explain why order_id should stay text.
4. Calculate each order amount and the all-orders total. Keep the row calculations visible so you can locate an arithmetic error.
5. Calculate known-paid, recorded-unpaid and unknown-status amounts. Count the records in each category too.
6. Reconcile the category counts and amounts to the whole table. Explain what the check proves and what it does not prove.
7. A colleague changes order 0013's missing status to FALSE and says, "The total is unchanged, so the change is harmless." Explain the problem in one or two sentences.

A hint if needed

Start with order 0012: four items at 12.50 each. For the payment question, identify which rows are definitely TRUE, definitely FALSE or missing before calculating the categories.

Save your attempt and uncertainty before reading the review. An honest "I am unsure why this is text" is more useful than a copied answer. Use the review to repair the idea, then try the changed requirement on Changed retry.

Changed retry - after your first review

Changed task: answer a different question

The owner now asks: "What is the value of orders whose payment status is known, whether paid or unpaid?" Include TRUE and FALSE. Exclude missing. This differs from asking only for paid orders.

order_id	item	quantity	unit_price	ordered_on	is_paid
A015	Pencil	5	8.00	2026-09-05	TRUE
A016	Eraser	2	5.00	2026-09-05	FALSE

order_id	item	quantity	unit_price	ordered_on	is_paid
A017	Bag	1	300.00	2026-09-05	(missing)

State the row meaning and counts. Calculate the row amounts. Report the IDs, count and total for known status; report unknown status separately. Reconcile to the whole table. Explain why A017 is text and why missing does not mean unpaid.

Save an attempt before opening the review on Retry review. The separation supports honest practice; it is not a technical lock.

Revision in your own words

Remember	Meaning
Table	A grid of named fields and records
Record / row	One entry; always state what it represents
Field / column	One kind of information collected for each entry
Cell and value	A cell is a position; its value is the information there
Header	Names fields; it is not another data record
Type	The intended category of a value, not only its appearance
Identifier	A label such as 0007 or A017; text can preserve its identity
Missing	No supplied answer; not automatically zero or FALSE
Reconciliation	Compare results that should agree; remember the check's limits

What understanding looks like

You can read a new row aloud, identify fields and records, explain types, calculate an amount, preserve an unknown status and check the result. After a mistake, you can explain the correction and solve the changed requirement without copying the first answer.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-01 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-02

Grain: what one row represents

Watch first

1. [What is Data? An Introduction to Data Terminology and Concepts](#)

Reused visual from DE01-01: 00:00–06:38. Optional reminder about records and summaries. The book teaches the precise grain sentence and order-versus-line distinction.

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-01. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: What is Data? An Introduction to Data Terminology and Concepts, 00:00-06:38](#). Revisit only if needed for individual records versus summary data. Our book teaches the precise grain sentence and order-versus-line distinction. This is the same introduction already assigned in DE01-01. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

The job of this lesson

Before counting rows or adding money, say exactly what each row describes. Complete DE01-01 first. You already know rows, fields, quantities and prices. Here we deliberately change the shop's recording rule: an order can contain several product kinds.

Words before we use them

Word	Plain meaning	Example
Grain	The precise meaning of one row	one product line within an order

Order line One entry for a product kind on an order

two notebooks on line 1

Aggregate Combine several rows into a summary

add line amounts for one order

Order-level value A fact about the whole order one delivery fee for the order

Imagine an envelope containing two order cards. The envelope is order 0007. One card describes notebooks and another describes pens. Counting cards gives two lines; counting envelopes gives one order. Both counts can be correct, but they answer different questions.

Follow one order slowly

order_id	line_no	item	quantity	unit_price	delivery_fee
0007	1	Notebook	2	49.50	20.00
0007	2	Pen	3	10.00	20.00

Rule: one row describes one product line within one order. The fee is 20.00 for the entire order, repeated here to expose a mistake. It is not a separate fee for each line. All amounts use the same currency and exclude taxes and discounts.

Line 1 is worth $2 \times 49.50 = 99.00$. Line 2 is worth $3 \times 10.00 = 30.00$. The goods total is 129.00. Add the single order delivery fee of 20.00 to get 149.00. If you add delivery_fee down the displayed column, you get 40.00 and a wrong payable total of 169.00. Every multiplication was correct; the error came from treating an order-level value as though it belonged independently to each line.

State a contract before calculating

Write three short sentences: "Input grain: one product line per order. Output grain: one order. Add the goods amounts across lines and include the order fee once." These sentences are a contract: a clear rule your result must obey. They make an otherwise hidden assumption visible.

The repeated order_id does not automatically mean a duplicate mistake. It repeats because the order has two legitimate lines. Conversely, simply selecting a different fee value is not enough: if one line said 20 and another said 25, you would need to investigate which fee is correct. Counting once assumes you have established one consistent fee for the order.

Picture the result

```
order_id line_count goods_amount delivery_fee_once
payable
0007 2 129.00 20.00 149.00
```

The output has one row because it answers an order-level question. You have changed the grain intentionally. Later SQL grouping and joins will make this much easier at scale, but their correctness still depends on these same meanings.

If you feel lost

Cover the numbers and identify the things being counted: orders, product lines or physical items. Here there is one order, two lines and five physical items. Then uncover the prices and ask which value belongs to which thing. Do not begin with a formula just because a column contains numbers.

Try it yourself

Order A100 has two lines: 2 folders at 25.00 each, and 4 pens at 10.00 each. Its single delivery fee is 15.00, repeated on both input rows. State the input and output grains; count orders, lines and physical items; calculate goods and payable; explain the wrong result from summing both displayed fees.

Changed retry - after your first review

Order A101 has three lines: 1 bag at 200.00, 2 pencils at 8.00 and 1 ruler at 20.00. There is one 10.00 delivery fee. Calculate the goods and payable, then explain why a result with three rows would not yet be the requested one-row-per-order summary.

Revision in your own words

State the grain before counting, grouping or combining data. Rows, orders, lines and items are different counts. A repeated identifier can be legitimate at a finer grain. Summarise each value according to the thing it describes.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-02 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-03

Keys, duplicates and missing values

Watch first

[1. Data Accuracy and Validation: Methods to ensure quality](#)

00:00–10:29. Visual context for validity and missing values. The book adds candidate/composite keys, duplicates and missing-versus-zero reasoning.

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-02. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Data Accuracy and Validation: Methods to ensure quality, 00:00-10:29.](#)

[See valid](#), invalid, missing and unusual values. The book adds primary-key candidates, duplicate-record handling and why missing is not zero. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

The job of this lesson

Identify a row reliably and preserve the difference between a known answer and no answer. Complete DE01-02. You now know why `order_id` can repeat legitimately in an order-line table. We return first to the simpler rule: one row per order containing one product kind.

Words before we use them

Word Plain meaning Important limit

Key A field or combination intended to identify one row

check it at the declared grain

Unique No repeated value in the chosen set

uniqueness alone does not establish meaning

Duplicate A repeated item under a stated rule

define whether you mean whole row or key

Composite key Two or more fields used together

`order_id` plus `line_no`

NULL / missing A value has not been supplied or is unknown

not automatically zero or FALSE

A shop's customer name is often a poor order key: the same person can place several orders and different people can share a name. A key should identify the intended record, not merely look distinctive in a tiny sample.

Inspect a deliberately damaged delivery

source_row	order_id	item	quantity	is_paid
1	0007	Notebook	2	TRUE
2	0008	Pen	3	FALSE
3	0008	Pen	4	FALSE
4	(missing)	Folder	1	TRUE
5	0010	Marker	2	(missing)

The declared grain is one row per order. `order_id` is intended to be the key. Rows 2 and 3 share 0008 but disagree about quantity. They are a key conflict, not identical whole rows. Deleting either one merely because it appears second would choose a quantity without evidence. Row 4 lacks the key needed to identify the order. Row 5 has an ID but lacks payment status. These are different issues and deserve different responses.

Follow with me once

Count the physical rows: five. List supplied IDs: 0007, 0008, 0008, 0010. There are three different supplied IDs. This does not prove there are exactly three real orders in the delivery; the row with the missing ID could be a fourth order or could relate to another one. The conflicting 0008 rows also need investigation.

Keep the original delivery. Make an issue record saying: "0008 appears twice with quantities 3 and 4; source confirmation required." Record the missing ID separately. For 0010, preserve an unknown status. Do not invent an ID or set the status to FALSE to make a completeness check look better.

Now return to the two-line order in DE01-02. `order_id` alone repeats legitimately. The pair (0007,1) identifies the notebook line and (0007,2) identifies the pen line. The proposed composite key is `order_id` plus `line_no`. Its usefulness depends on checking that each pair occurs once and neither component is missing.

Why missing is not an ordinary value

Quantity zero could mean a measured count of none. Missing quantity means no count was supplied. FALSE under `is_paid` is a supplied unpaid status; a blank means you do not know. Excel blanks, text such as "NULL", and SQL NULL are different representations. We discuss their shared meaning here, but do not pretend they behave identically in formulas. In our workbook a genuinely empty status cell is unknown; the printed phrase (missing) is only a reading aid.

Try it yourself

At one-row-per-order grain you receive IDs A1, A2, A2 and a missing ID. The two A2 rows have amounts 20.00 and 25.00. State the number of source rows, different supplied IDs, key-conflict group and missing-key rows. Explain what you may conclude and what needs confirmation.

Changed retry - after your first review

At one-row-per-order-line grain, the pairs are (A1,1), (A1,2), (A2,1), (A2,1). Identify the duplicated composite key and explain why repeated A1 alone is not a fault. One A1 status is FALSE and the other is missing: can both be called unpaid?

Revision in your own words

Define a key at the grain you intend. Check missing key parts and repeated keys. Distinguish identical rows from conflicting rows. Preserve unknown values and report uncertainty instead of inventing a convenient answer.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-03 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-04

Data quality and useful validation

Watch first

[1. Data Accuracy and Validation: Methods to ensure quality](#)

Reused visual from DE01-03: 00:00–10:29. No second mandatory watch. The book teaches how to design and evaluate useful quality checks.

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-03. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Data Accuracy and Validation: Methods to ensure quality, 00:00-10:29.](#)

[Reuse the](#) data-validation visual from DE01-03 if needed. Apply our own quality questions and preserve evidence; no second mandatory watch. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

The job of this lesson

Design checks that can catch specific mistakes, and understand why a passing check is limited evidence. Complete DE01-03. Validation means testing data against a stated rule. It is useful only when the rule matches what the data is supposed to mean.

Words before we use them

Word Plain meaning A shop example

Completeness Required information has been supplied

every order has an ID

Validity A value follows an allowed rule quantity is a whole number above zero

Uniqueness An intended key does not repeat

one order_id per order

Consistency Related facts do not contradict a shared rule

categories add to the full total

Accuracy A recorded fact matches reality the paid status matches payment evidence

Timeliness Data is recent enough for the question

today's stock is not last month's stock

These are practical engineering checks, not a universal list that every organisation names identically. A source-quality video from Statistics Canada discusses a different statistical framework, including relevance, interpretability and accessibility. Its purpose is to judge whether

a source suits a decision. Our row checks examine concrete rules within an input. Keep those purposes distinct.

Follow with me once

Use the four clean orders from DE01-01. The shop states: one row per order; every ID required; quantities are positive whole numbers; prices are nonnegative; paid status may be unknown. Check each rule separately.

Check	Observed result	What it supports
Row count	4	matches this delivery's stated size
Missing IDs	0	every source row has an identifier
Repeated IDs	0	no key repeats in this sample
Positive whole quantities	4 of 4	quantities follow the stated rule
Missing paid status	1	one payment answer remains unknown
Category amount sum	$124 + 30 + 60 = 214$	the three groups cover the total consistently

The missing status is not automatically a rejection: the rule allows unknown. It should still be reported because the owner cannot count that order as definitely paid or unpaid. A requirement and an observation are different. First decide what is allowed; then measure what you have.

A passing total can hide an error

Change order 0010 from missing to FALSE. The total remains 214.00, but recorded unpaid becomes 90.00 instead of 30.00. A total check passes while the classification becomes wrong. Add a check that compares the payment category with the original status. Likewise, a correctly formatted date can still be the wrong real purchase date. Validity cannot establish accuracy without independent evidence.

Build a small issue report

For each failed rule, write the rule, affected row or key, observed value, expected condition and proposed action. For example: "Quantity rule: order A20 has -2; positive whole quantity required; preserve source and request confirmation." Do not silently change -2 to 2. It might represent a return under a different business rule, or it might be a typing error. The owner must establish the intended meaning.

If you are unsure which check to choose

Ask "What mistake am I trying to detect?" A duplicate-key check targets repeated IDs. A row-count comparison targets lost or extra records. A source-to-result comparison targets changed values. A freshness check targets old data. Use several checks when the risks differ, and state what remains unverified.

Try it yourself

You expect 3 one-row-per-order records. Receive A10 quantity 2 price 10.00; A11 quantity -1 price 20.00; A11 quantity 3 price 5.00. All payment statuses are missing and unknown is allowed. Identify the passing row-count check, failed key check, failed quantity check and allowed-but-important missing statuses. State a responsible next action.

Changed retry - after your first review

A delivery has 4 rows, 4 unique IDs, valid positive quantities and category totals that reconcile. The source date is a month old, but the owner asks for today's available stock. Explain why you still cannot claim it answers the question accurately.

Revision in your own words

State the rule; inspect the value; record the result and its limit. A pass means a particular check passed. It does not prove all data is correct. Preserve source, report unknowns and investigate conflicting evidence.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-04 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-05

Meet Excel: workbook, worksheet, cells and formulas

Watch first

[1. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

01:15-04:09

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-04. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 01:15-04:09.](#) [Start Excel and](#) identify the screen areas. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

The job of this lesson

Find your way around Excel without guessing which sheet or cell a lesson refers to. Complete DE01-04 and confirm spreadsheet access in M00. Use labs/M01/M01_Practice.xlsx. Make a copy under your practice folder before editing. The file has no macros.

Words before we use them

Word	Plain meaning	In this workbook
Workbook	The whole spreadsheet file	M01_Practice.xlsx
Worksheet	One grid inside the workbook	Orders or Raw

Cell address Column letter followed by row number

C3

Range A group of cells A1:F5

Formula bar The area showing a selected cell's contents
often beside an fx label

Formula An instruction beginning with = =2*49.5

A workbook is like a binder; worksheets are its pages. C3 is a position on one page. C3 on Orders and C3 on Independent are different places, even though their short addresses match. Always check the sheet name before typing.

Follow with me once

Open your copied workbook. If Excel opens a downloaded file in Protected View, inspect that it is this expected course file before enabling editing. Do not enable macros for this batch; none are needed. Select the Orders sheet tab. The header occupies A1:F1. Click A2: you should see the text identifier 0007. Click C3: its quantity is 3. Click D3: its unit price is 10.00. Read the Name Box, usually left of the formula bar, to confirm each address.

Click A1 and drag to F5 to select the complete small table. Alternatively, type A1:F5 into the Name Box and press Enter. A colon means every cell in that rectangular range. It includes one header row and four records. The selected range therefore has five spreadsheet rows while the dataset has four records. This is the same distinction you learned in DE01-01.

Click an empty cell J2. Type $=2*49.5$ and press Enter. The asterisk means multiply; you should see 99. Select J2 again. The formula bar should show the instruction $=2*49.5$ even if the cell displays 99 or 99.00. One is the stored formula; the other is its result. Save the workbook using Ctrl+S, close it and reopen it from its copied location. Return to Orders!J2 and check the result and formula again.

Understand navigation and selection

Clicking a cell selects it. Typing replaces its current contents, so inspect the address first. Press Esc to cancel an edit before committing it. Ctrl+Z can undo a recent accidental change. Double-clicking or using the formula bar lets you edit existing contents; simply typing in a selected cell usually replaces them. Experiment only in your working copy.

The ribbon groups commands under tabs such as Home, Insert and Data. The bottom sheet tabs choose grids. The formula bar shows the current cell's contents. These areas serve different jobs. A video might show a differently coloured ribbon or a different number of visible buttons. Find the named action and confirm its effect instead of matching every pixel.

If something goes wrong

If you see ##### in a date or number cell, try widening the column; the underlying value may be fine. If a formula displays as text, check that it begins with =, lacks a leading apostrophe, and is not entered in a Text-formatted cell. Change a practice cell to General and re-enter the formula if necessary. If every formula is displayed instead of results, inspect the Show Formulas setting under Formulas. Do not assume all these symptoms have the same cause.

Try it yourself

On Orders, identify the contents of B4, C2 and F5. In J3 calculate 3 multiplied by 10 using a formula. Save and reopen. Record the sheet, cell address, displayed result and formula. Explain why row 1 is not an order.

Changed retry - after your first review

On Independent, find C3 and D3 without copying the Orders values. In J2 multiply the two values using a formula, then inspect the formula bar. Keep this result separate from Orders!J2.

Revision in your own words

Workbook means file; worksheet means one grid. Check sheet and address before typing. A1:F5 is a range. A formula begins with =; a displayed result is not the formula text. Save, close and reopen to verify your working file.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-05 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-06

Enter, save and structure tabular data safely

Watch first

[1. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

04:09–07:35

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-05. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 04:09-07:35.](#) [Watch a sales](#) tracker being entered; compare its layout with our different source. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

The job of this lesson

Create a small editable table without destroying identifiers or mixing values with decoration. Complete DE01-05. Work in your copied practice workbook. Raw preserves the provided text delivery. Orders is the working interpretation with quantities and prices stored as numbers, dates as dates, paid answers as Booleans and missing status as an empty cell.

Words before we use them

Word	Plain meaning	Example
Stored value	What a cell actually contains	numeric 10

Number format How Excel displays a stored value
showing 10 as 10.00

Text identifier	A label preserved as characters	0007
Boolean	A true-or-false value	TRUE or FALSE

Import Bring a file's contents into an application
read a CSV into columns

Making a cell look like an identifier is not always the same as preserving the original identifier. A custom number format might display numeric 7 as 0007. The visible result looks right, but the stored value is still 7. For our IDs, use text because the characters themselves are the label.

Follow with me once

Create a new worksheet using the plus button beside the tabs and name it Entry. Type these six headers into A1:F1: order_id, item, quantity, unit_price, ordered_on, is_paid. One header row is enough. Do not merge cells or insert a title row inside the data rectangle. A separate note beside the table can describe the source without interrupting the records.

Before entering IDs, select A2:A5 and use Home's number-format list to choose Text. Enter 0007, 0008, 0009 and 0010. Inspect A2 in the formula bar: the leading zeros should remain. If you already entered numeric 7 and lost the zeros, changing its format to Text cannot recover characters that were discarded. Return to Raw and re-enter the source ID after choosing Text.

Enter the items, quantities and prices from DE01-01 into B2:D5. Quantities and prices must be numbers, so do not type currency words into them. Format D2:D5 to show two decimal places.

Enter the first date with =DATE(2026,9,1). DATE constructs a date from year, month and day, avoiding uncertainty over whether 01/09 means January 9 or September 1. Use that date for rows 2 and 3; use =DATE(2026,9,2) for rows 4 and 5. Format E2:E5 as yyyy-mm-dd if available through Format Cells, Custom. Excel stores dates as date serial numbers, then formats their display.

In F2 type TRUE, in F3 FALSE and in F4 TRUE without quotation marks. Leave F5 genuinely empty. Do not type the words (missing) into it: that phrase is only the printed book's marker. In J2 enter =ISTEXT(A2), J3 =ISNUMBER(C2), J4 =ISNUMBER(E2), and J5 =ISLOGICAL(F2). These checks ask whether the stored cell value is text, numeric, numeric/date, or Boolean. Each should return TRUE. ISNUMBER(E2) alone does not prove that the number is the intended date; also inspect its displayed date and source.

Preserve the complete file

Save as an XLSX workbook to retain sheets and formulas. If you deliberately export CSV, understand that the export is a simpler data file, not a replacement for the complete workbook. Keep the XLSX. Close and reopen it, check the IDs and dates, and compare row count with Raw. Do not overwrite Raw with your edited entry table.

If something differs

Some regional Excel settings use semicolons between formula arguments. If =DATE(2026,9,1) is rejected, try =DATE(2026;9;1) and record your locale setting. If a numeric calculation ignores a quantity, inspect whether it was stored as text. If a blank status appears as zero after a formula or import, return to the source and inspect the transformation; do not conclude the source supplied a zero.

Try it yourself

Create a new sheet EntryAttempt with two records: 0020, Ruler, 2, 20.00, 2026-09-06, TRUE; and 0021, Bag, 1, 250.00, 2026-09-06, missing status. Preserve text IDs and an actually empty final status cell. Check the intended types, save and reopen. Explain which values are stored and which features are only display formatting.

Changed retry - after your first review

Add order A022, Pencil, 3, 8.00, 2026-09-07, FALSE to EntryAttempt. Explain why the ID type still works and why this supplied FALSE must remain distinct from the previous blank.

Revision in your own words

Choose intended types before entry or import. Keep IDs as text, numeric measures as numbers, unambiguous dates and real Boolean answers. Leave missing cells missing. Formatting changes presentation. Preserve Raw and keep the full XLSX workbook.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-06 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-07

Sort and filter without damaging records

Watch first

[1. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

11:37-13:05

[2. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

13:57-15:46

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-06. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 11:37-13:05. Watch filtering.](#) Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 13:57-15:46. Watch](#) whole-record sorting. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

The job of this lesson

Change which records you see and their order without separating an ID from its values. Complete DE01-06. Make a fresh copy of the Orders sheet and call it SortPractice. In desktop Excel, right-click the sheet tab, choose Move or Copy, select Create a copy, and rename the copy. Work only on that copy.

Words before we use them

Word Plain meaning Effect
Sort Rearrange rows by a chosen field
put high prices first

Ascending	Small to large or A to Z	1, 2, 3
Descending	Large to small or Z to A	3, 2, 1

Filter Display only rows meeting a condition
show quantity 2

Clear filter Display the hidden rows again it does not restore deleted rows

A record is a connected set of values. Moving only its price while leaving the ID behind creates a different claim about the order. Think of sorting whole order cards, not cutting out the price labels and shuffling those separately.

Follow with me once

Select A1:F5 on SortPractice. Use Data, Sort. Confirm My data has headers. Choose unit_price, Cell Values, Largest to Smallest. If Excel asks whether to expand the selection, the intended selection must include all six fields. Review the whole range rather than approving a sort of column D alone.

The expected order IDs are 0007, 0010, 0009, 0008 because their unit prices are 49.50, 30.00, 25.00 and 10.00. Check two whole records: 0010 must still describe Marker, quantity 2, price 30.00 and unknown status; 0008 must still describe Pen, quantity 3, price 10.00 and FALSE. A

sorted price column can look tidy while these associations are broken, so checking prices alone is insufficient.

Select the complete table and use Data, Filter. Open the quantity header's arrow, clear Select All and select only 2. Confirm the filter. Two records should remain visible: 0007 and 0010. Other rows are hidden by the filter, not deleted. Open the quantity filter and choose Clear Filter From quantity, or use Data's Clear action for filters. All four records should return, still in the sorted order.

Sort and filter answer different questions

Sorting asks which record comes first. Filtering asks which records qualify for display. Clearing a filter does not undo a sort. Neither action should change a stored quantity or price. A simple SUM over a normal range may still include filtered-out rows; do not assume every total follows visible rows. Later, use an explicitly appropriate filtered total or calculate the selected records independently. For this lesson, count the visible IDs directly.

If the result surprises you

If only the price column moved, use Undo immediately in your copy and compare with Raw. Recreate the copy if you are unsure which records were damaged. If fewer than four records appear after you clear one filter, another field may still be filtered. Inspect header funnel icons or clear all filters. If 100 sorts before 20 in an ascending list, the values may be text; inspect types before converting them.

Try it yourself

From a fresh four-order copy, sort by quantity descending, then by order_id ascending to break equal-quantity ties. Record the expected ID order. Filter item to Pen, record its complete row, then clear the filter and confirm all four records remain.

Changed retry - after your first review

Use a fresh copy of Independent. Filter quantity to values greater than 1. List the IDs and explain whether any source record was deleted. Clear the filter and verify the full count.

Revision in your own words

Sort complete records. Check at least one ID-to-value association after sorting. Filters hide; they do not delete. Clear filters to inspect the full input. A visible subset and a whole-range formula can cover different records.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-07 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-08

Basic formulas and essential functions

Watch first

[1. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

15:46–21:01

[2. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

21:29–22:39

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-07. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 15:46-21:01. Watch formula](#) totals. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 21:29-22:39. Watch an](#) average. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

The job of this lesson

Turn the arithmetic from DE01-01 into formulas that respond to the source cells. Complete DE01-07. Use the unsorted Orders sheet with the original four records on rows 2-5. In this lesson all amounts are in the same currency and there are no taxes, discounts or delivery fees.

Words before we use them

Word	Plain meaning	Example
Reference	A formula's address for an input	C2

Function A named calculation provided by Excel

SUM

Argument An input supplied inside parentheses

G2:G5 in SUM(G2:G5)

Criterion A condition selecting values "Paid"

Recalculate Update results after input changes

quantity changes, amount follows

Build the amount one row at a time

Type amount in G1. In G2 type `=C2*D2`. This reads quantity from C2 and unit price from D2, then multiplies them. It should show 99.00. The formula `=2*49.5` gives the same result today, but it does not point at the cells. If quantity changes, a reference-based formula updates; hard-coded numbers stay as you typed them.

Copy G2 to G3:G5, or select G2:G5 and use Ctrl+D to fill down. Inspect G3 in the formula bar: it should be `=C3*D3`. The row numbers moved to match the next order. Results should be 99.00, 30.00, 25.00 and 60.00. Format G2:G5 to two decimals for readability.

In J7 type All orders and in K7 type `=SUM(G2:G5)`. SUM adds the four amounts. K7 should be 214.00. In J8 type Mean order amount and K8 type `=AVERAGE(G2:G5)`. AVERAGE divides the sum by the count of numeric amounts: $214 / 4 = 53.50$. In K9 type `=COUNT(C2:C5)`, giving four numeric quantities. In K10 type `=COUNTA(A2:A5)`, giving four nonempty ID cells. COUNT is for numeric values; COUNTA includes nonempty text such as our IDs. Do not interchange them just because both happen to return four here.

Classify payment without erasing unknowns

Type payment_group in H1. In H2 type:

```
=IF(ISBLANK(F2),"Unknown",IF(F2=TRUE,"Paid","Unpaid"))
```

Read it from the inside checks outward: ISBLANK(F2) asks whether the source cell is truly empty. IF uses the first result when the condition is true and the second when it is false. If F2 is empty, return Unknown. Otherwise inspect the supplied Boolean: TRUE gives Paid; the remaining permitted Boolean FALSE gives Unpaid. This formula assumes you already checked that nonempty status cells really contain Booleans. Arbitrary text such as "maybe" must not be accepted as unpaid.

Fill H2 down to H5. The categories should be Paid, Unpaid, Paid, Unknown. In K12 use `=SUMIF(H2:H5,"Paid",G2:G5)`. SUMIF selects rows where the category equals Paid, then adds the matching amount cells. It gives 124.00. Use corresponding formulas for Unpaid in K13 and Unknown in K14: 30.00 and 60.00. In K15 use `=SUM(K12:K14)-K7`. Zero means the category amounts reconcile to the whole total. It does not prove the payment source is accurate.

If a formula fails

If you get #NAME?, check function spelling and quotation marks around text. If you get #VALUE!, inspect the input types rather than replacing the error with zero. If a formula is rejected at the commas, your locale may use semicolons. If the total is 114.50, you added unit prices instead of row amounts. If missing is grouped as unpaid, inspect the blank test first.

Try it yourself

On Independent, calculate amount and payment_group with formulas. Report the four amounts, whole total, mean amount, paid/unpaid/unknown totals, numeric quantity count and a reconciliation difference. Change the first quantity temporarily from 1 to 2, inspect the total, then restore 1 and save the original result. Preserve both observations in your attempt.

Changed retry - after your first review

On Retry, calculate all three order amounts. The question is now the total with a known payment status, including both paid and unpaid. Report the selected IDs, their total and unknown separately. Use your amount and payment_group columns to support the answer.

Revision in your own words

Use references for changing inputs. SUM adds; AVERAGE calculates a mean; COUNT counts numbers; COUNTA counts nonempty cells. IF chooses by a condition. SUMIF adds matching rows. Protect unknowns, inspect formulas and change one input to check that the result responds.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-08 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-09

Relative and absolute references

Watch first

[1. Excel Cell References: Relative, Absolute and Mixed](#)

00:00-03:49

[2. Excel Cell References: Relative, Absolute and Mixed](#)

05:06-06:57

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-08. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Excel Cell References: Relative, Absolute and Mixed, 00:00-03:49.](#)

[Watch how](#) relative references move and how dollar signs fix a reference. Stop manually at the listed end. Public title, creator, runtime and published chapter boundaries inspected; full playback not verified.

[Instructor demonstration: Excel Cell References: Relative, Absolute and Mixed, 05:06-06:57.](#)

[Reinforce](#) absolute references using the instructor's different business example. Stop manually at the listed end. Public title, creator, runtime and published chapter boundaries inspected; full playback not verified.

The job of this lesson

Copy a formula while keeping a shared assumption fixed. Complete DE01-08. Use Orders with amount in G2:G5. We add a fictional 10% discount purely to practise references. This is not a tax rule or a claim about the shop's real prices.

Words before we use them

Word	Plain meaning	Example
Relative reference	Moves with a copied formula	G2 becomes G3 one row down

Absolute reference Keeps the specified row and column fixed when copied

\$K\$2

Mixed reference Fixes only row or column K\$2 or \$K2

Assumption An input rule used in a calculation

the chosen discount rate

Percentage A number expressed per hundred

10% equals 0.10

Copying a relative reference is like saying "use the amount on this row." An absolute reference is like saying "always use this particular settings cell." Both are useful; adding dollar signs everywhere is not automatically safer.

Follow with me once

On Orders, put Discount rate in J2 and 10% in K2. Type after_discount in I1. In I2 enter $=G2*(1-K\$2)$. The 1 means 100% of the amount. Subtracting 0.10 leaves 0.90, so the result is $99.00 \times 0.90 = 89.10$. The dollar signs are reference markers, not a currency instruction.

Copy I2 down through I5. Inspect each formula. G2 should become G3, G4 and G5, but $\$K\2 should remain unchanged. The results should be 89.10, 27.00, 22.50 and 54.00. Their total is 192.60. Every row uses its own amount and the one shared rate.

Now make a temporary experiment in a spare area. Enter $=G2*(1-K2)$ in M2 and fill down to M5. If K3:K5 are empty, the later copied formulas refer to those empty cells instead of the rate. They may leave later amounts undiscounted, giving plausible-looking but inconsistent results. Inspect M3's formula: it points at K3. Restore or clear the experiment after recording the explanation. Do not change empty cells to 10% just to conceal a wrong reference pattern.

Test the assumption honestly

Change K2 from 10% to 20%. All after_discount rows should update: 79.20, 24.00, 20.00 and 48.00, totalling 171.20. Restore 10%. This is stronger evidence than checking only I2, because a copying error often leaves the first row correct and damages later rows.

Mixed references matter when copying across as well as down. $\$K2$ fixes column K but permits the row to move. $K\$2$ fixes row 2 but permits the column to move. You only need to recognise this now; our one-rate exercise needs both locked. In desktop Excel, F4 can cycle a selected formula reference through variants; some laptops require $Fn+F4$. You may type the dollar signs directly.

If the result is wrong

If an amount becomes negative, check whether you entered 10 instead of 10%. Ten is 1000%, not 10%. If only the first row is discounted, inspect the copied rate reference. If every row shows 89.10, you probably fixed G2 as well as the rate. The order amount should move by row; the rate should stay fixed.

Try it yourself

On Independent, put a shared discount of 5% in K2 and calculate each discounted amount in I2:I5. Inspect the final copied formula and total. Temporarily change the rate to 15%, report the new total, then restore 5%.

Changed retry - after your first review

On Retry, apply a shared 12% discount to the three row amounts. State which part of the formula must move and which must remain fixed. Check the total after changing the shared rate to 0%, then restore 12%.

Revision in your own words

Relative references move when copied. Dollar signs lock the following column or row. Lock shared assumptions, let row-specific amounts move, and inspect the last copied formula. Test another input value before trusting a copied column.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-09 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-10

Excel Tables and structured working habits

Watch first

[1. Excel Cell References: Relative, Absolute and Mixed](#)

08:10-09:55

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-09. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Excel Cell References: Relative, Absolute and Mixed, 08:10-09:55.](#) [Watch](#) structured references. The book supplies the full Ctrl+T creation and source-growth exercise. Stop manually at the listed end. Public title, creator, runtime and published chapter boundaries inspected; full playback not verified.

The job of this lesson

Give a growing data range a clear boundary and readable column names. Complete DE01-09. A general table is rows and fields, as in DE01-01. An Excel Table is a particular workbook feature that manages a rectangular dataset. Those meanings are related but not identical.

Words before we use them

Word	Plain meaning	Example
------	---------------	---------

Excel Table	A named data object inside a worksheet	
-------------	--	--

tblOrders		
-----------	--	--

Structured reference	A formula referring to a table's named column	
----------------------	---	--

tblOrders[amount]		
-------------------	--	--

Calculated column	A table column using the same row rule	
-------------------	--	--

quantity multiplied by unit_price		
-----------------------------------	--	--

Total row	A special summary row managed by the Table	
-----------	--	--

a visible total below its records		
-----------------------------------	--	--

Follow with me once

Make a fresh copy of Orders, call it TablePractice and keep only the six source fields and your amount/payment_group columns in A1:H5. Notes and assumptions should remain outside that rectangle. Select A1:H5, press Ctrl+T, check My table has headers and confirm. Click inside the new Table. On Table Design, change the Table Name to tblOrders. A valid name has no spaces and should be distinct within the workbook.

Filter arrows and banded formatting may appear. Those are useful visual cues, but the important change is that Excel now knows the dataset's boundaries and named fields. In the first amount data cell, enter `=[@quantity]*[@unit_price]`. The @ means this row. The formula reads this record's quantity and unit_price rather than a fixed record from somewhere else. Excel normally fills the calculated column; inspect the remaining rows and use fill-down if your settings do not do so automatically. Expected amounts remain 99, 30, 25 and 60.

In J7 enter Table total and in K7 enter `=SUM(tblOrders[amount])`. This names the amount column rather than the particular range G2:G5. Now add a fifth record immediately below the Table: 0015, Pencil, quantity 5, unit_price 8.00, date 2026-09-03, TRUE. The Table should

expand to include it, and its amount should be 40.00. Check Table Design, Resize Table if it did not expand. The structured total should become 254.00. Do not assume expansion occurred simply because a coloured row appeared nearby.

Compare with a fixed range

A formula `=SUM(G2:G5)` still describes those original four amount cells. Adding a fifth record on row 6 can leave that fixed range behind. A structured reference can follow the Table's data column when the Table actually expands. This helps prevent one common mistake, but it does not validate the new record's types, key or business meaning. A duplicated ID inside a perfectly formatted Table is still duplicated.

If you enable the Table's Total Row, inspect the formula it uses. Excel commonly inserts SUBTOTAL for filtered-aware summaries. SUM and SUBTOTAL can intentionally cover different visible/all-row sets. For now, keep your explicit whole-table SUM in K7 and clear filters when comparing totals. We are learning a named data structure, not every Table option at once.

If the result differs

If headers become Column1, Column2 and so on, you may have omitted the header checkbox or selected the wrong range. Undo and recreate the Table from A1:H5 in a fresh copy. If a new amount stays blank, inspect whether the row belongs to the Table and whether the calculated-column formula filled. If the total is 214 after adding the new order, inspect its reference and Table boundary.

Try it yourself

Create an Excel Table from Independent's six fields plus calculated amount and payment_group columns. Name it tblIndependent. Use a structured reference for its whole amount total. Add 0016, Eraser, quantity 2, price 5.00, date 2026-09-05, FALSE. Record the new row count, added amount and total. Preserve the original Independent sheet or use a copy.

Changed retry - after your first review

In the Table copy, change the newly added Eraser quantity from 2 to 3. Predict the new total before looking at it. Then restore quantity 2 and save.

Revision in your own words

An Excel Table gives data named boundaries. Structured references name its columns. @ means this row. Inspect expansion and calculated columns. A managed rectangle can still contain wrong data, so retain key, type and reconciliation checks.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-10 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-11

Practical cleaning and exact lookups

Watch first

[1. How to use VLOOKUP in Excel](#)

01:20-03:40

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-10. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: How to use VLOOKUP in Excel, 01:20-03:40. Watch VLOOKUP attaching a matching](#) value. In our lab use FALSE for exact match and keep unexpected #N/A visible. Stop before IFERROR; our cleaning steps are taught in writing. Stop manually at the listed end. Public title, creator, runtime and YouTube-displayed chapter boundaries inspected; full playback not verified.

The job of this lesson

Clean a specific text problem without erasing source evidence, then attach a category using an exact match. Complete DE01-10. Use the Dirty and Lookup sheets. Do not change Raw. A lookup finds a matching value in another table; it does not establish that the match is a valid business rule by itself.

Words before we use them

Word Plain meaning Example

Whitespace Spaces or similar invisible characters

a trailing space after Pen

Normalise Apply an agreed consistent representation

trim ordinary outside spaces

Lookup table A small table mapping labels to information
item to category

Exact match Require the lookup label to match

Pen must match Pen

Unmatched No matching lookup entry was found

Ruler is absent from this map

Inspect before changing

Dirty has four rows: Notebook, " Pen ", Folder and Ruler. The quotation marks in this sentence expose ordinary spaces; they are not part of the cell contents. Lookup maps Notebook to Paper, Pen to Writing, Folder to Paper and Marker to Writing. The course rule is to trim ordinary outside spaces from item labels and keep the unmodified item in its source column. It does not authorise correcting arbitrary spellings or inventing categories.

In Dirty, type item_clean in C1. In C2 enter =TRIM(B2) and fill down. TRIM removes leading/trailing ordinary spaces and reduces runs of ordinary internal spaces to one. It is not a universal cleaner for every Unicode whitespace character. Inspect B3 and C3 in the formula bar: the source contains spaces around Pen; the clean result does not. Keep both columns so the change is visible.

Attach a category with a readable exact lookup

Type category in D1. In D2 enter:

```
=VLOOKUP(C2,Lookup!$A$2:$B$5,2,FALSE)
```

VLOOKUP searches the first column of the supplied table. C2 is the cleaned item to find. Lookup! names the other worksheet; the exclamation mark separates its name from the range. \$A\$2:\$B\$5 fixes the lookup rectangle when the formula is copied. The 2 asks for the second column, category. FALSE explicitly requests an exact match; here it is a lookup option, not the paid status from our order table.

Fill down to D5. The expected categories are Paper, Writing, Paper and #N/A. Ruler is not in Lookup, so #N/A is useful evidence of an unmatched label. Do not replace it with Paper just because a ruler feels related to stationery. The mapping rule belongs to the source owner. Record an unmatched issue while retaining the row.

Do not hide unexpected errors

IFERROR can replace many errors with a chosen result, but a broad replacement can conceal a bad range or broken formula as well as a legitimate missing match. First examine the actual error. For this lesson, leave #N/A visible in the lookup practice until you have explained it. In a user report you could label a confirmed no-match case "Unmapped", with the original row retained, but that is a deliberate reporting choice rather than evidence that the problem vanished.

Also check the lookup's own key: each item should appear once. If Pen appears twice with different categories, choosing the first match would hide conflicting mapping instructions. Duplicate mapping keys require investigation. A lookup is not the same as a general database join, which we will study later.

If a known item fails to match

Inspect source and clean text, the lookup spelling and the selected range. Check that you fixed the lookup range when filling down. Do not enable approximate matching merely to make #N/A disappear: an approximate result can be a different category than intended.

Try it yourself

On a new sheet, create source items " Marker ", "Notebook" and "Ruler". Keep the originals, create a trimmed column and use the same fixed exact lookup. Report the two mapped results and the unmatched case. Explain the purpose of FALSE and the dollar signs.

Changed retry - after your first review

In a copy of Lookup, deliberately add another Pen row mapped to Paper while the original Pen maps to Writing. Without accepting a first-match result as correct, describe the conflict and the evidence needed to repair it.

Revision in your own words

Inspect before cleaning, state the permitted transformation and preserve the source column. Match exactly, fix the mapping range and check its keys. Unmatched or conflicting mappings are issues to explain, not invitations to guess.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-11 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-12

PivotTables and Power Query awareness

Watch first

[1. Excel Pivot Tables Tutorial for Beginners](#)

01:24-03:40

[2. Excel Pivot Tables Tutorial for Beginners](#)

09:15-09:49

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-11. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Excel Pivot Tables Tutorial for Beginners, 01:24-03:40. Watch creating a PivotTable](#) and summarising sales by a category. Our exact fields and totals differ. Stop manually at the listed end. Public title, creator, runtime and published chapter boundaries inspected; full playback not verified.

[Instructor demonstration: Excel Pivot Tables Tutorial for Beginners, 09:15-09:49. See why a PivotTable needs](#) refresh. Power Query is awareness-only in this lesson; no Power Query clicks are required to pass this stage. Stop manually at the listed end. Public title, creator, runtime and published chapter boundaries inspected; full playback not verified.

The job of this lesson

Produce a small category summary and recognise the difference between summarising data and preparing it. Complete DE01-11. The required practical task is a PivotTable in Excel. Power Query is introduced as a repeatable preparation tool; its additional hands-on branch is optional if your installed edition exposes it. Do not claim a missing feature has been practised.

Words before we use them

Word Plain meaning Example

PivotTable An Excel feature that groups and summarises source rows

amount by item category

Rows area The PivotTable setting for group labels

Paper and Writing

Values area The setting for the measured summary

Sum of amount

Refresh Update a derived result from its source

rebuild a pivot after an edit

Power Query A tool for importing and applying repeatable preparation steps

trim a column, then reload

Prepare the known source

Use the supplied PivotSource sheet. It contains the four original order IDs and amounts, with the exact approved categories from Lookup: Notebook and Folder are Paper; Pen and Marker are Writing. Its grain is still one row per order. Expected total is 214.00. The provided workbook intentionally contains a plain source table and a formula-based reference summary; you create the native PivotTable yourself in this task.

Follow with me once

Click within PivotSource's A1:D5 range. Use Insert, PivotTable, From Table/Range if asked. Confirm that A1:D5 is the source and choose New Worksheet. In the fields panel, drag category into Rows and amount into Values. Check that Values says Sum of amount, not Count of amount. If it says Count, inspect whether the source amounts are stored as text; do not merely rename the heading to Sum.

You should see Paper 124.00 and Writing 90.00, with Grand Total 214.00. Paper combines 99.00 for Notebook and 25.00 for Folder. Writing combines 30.00 for Pen and 60.00 for Marker. The summary grain is one row per category. Four input records becoming two group rows is intentional; it is not evidence that two orders were lost.

Rename the result sheet PivotAttempt. Change Marker's source amount temporarily from 60.00 to 70.00. Before refreshing, inspect whether your PivotTable still shows the old result. Refresh it through PivotTable Analyse, Refresh or right-click, Refresh. Writing should become 100.00 and Grand Total 224.00. Restore 60.00 in the source and refresh again. Save the workbook with the original 214.00 total. A source edit and a refreshed summary are separate events.

What Power Query adds

Our TRIM formula produced a cleaned column. Power Query can record preparation steps, then apply them again when refreshed against a source. For example: read the Dirty source; keep its original item label; duplicate that column; trim the duplicate; load the result into a new table. Each step has an order. A later step can depend on a column name created earlier, so renaming or removing it can break later steps.

If desktop Excel shows Data, From Table/Range, try this on a copy of the two-column Dirty source. Make it an Excel Table first. In the query editor, duplicate the item column and apply Transform, Format, Trim to the copy. Use Close & Load to a new worksheet. Inspect original and cleaned text. Do not perform this branch if the interface or licence does not provide the feature; record it as awareness only and keep the required PivotTable task separate. The book explains the concept so missing optional software does not force random searching.

If the summary is wrong

Count of amount equal to 2 for each group is a count, not currency. Wrong group labels may come from incorrect mapping. A missing final source row can mean the selected source range did not include it. An old total after a source edit may need a refresh. Inspect the specific stage instead of repeatedly rebuilding the whole workbook.

Try it yourself

Create a small source with Paper amounts 60 and 40, Writing amount 50, and Bags amount 250. Build a PivotTable summing amounts by category. Record input count, output category count and total. Change the Paper 40 to 45, refresh and record the new Paper and grand totals; then restore and refresh.

Changed retry - after your first review

Using the same four-row source, create a PivotTable that counts records per category instead of adding amounts. Explain why its grand total has a different unit from 400.00.

Revision in your own words

PivotTables summarise by a chosen grain and function. Check source range, numeric types, aggregation and refresh. Power Query prepares data through recorded steps. Neither feature can repair an unclear business rule by itself.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-12 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-13

Chart choice and basic visual communication

Watch first

[1. Chandoo — Excel Tutorial for Beginners: Learn it FAST](#)

22:39-27:27

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-12. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Chandoo - Excel Tutorial for Beginners: Learn it FAST, 22:39-27:27.](#) [Watch chart](#) creation and adjustment. Stop manually at the listed end. Creator chapter titles and boundaries inspected 2026-09-13; full segment playback and transcript unavailable in this environment. Supplementary instructor demonstration.

The job of this lesson

Make a chart that answers one clear question without hiding its unit or overstating the evidence. Complete DE01-12. Use the category summary Paper 124.00, Writing 90.00. You need no artistic skill; the purpose is a readable comparison.

Words before we use them

Word	Plain meaning	Example
Category	A named group	Paper
Measure	The quantity being compared	order amount
Axis	A chart's scale or label direction	an amount scale starting at zero

- Bar chart Lengths compare category values
- Paper versus Writing
- Line chart Connected points usually show an ordered sequence
- daily totals across dates

A good chart begins with a question. "Which category has more order value in these four records?" calls for comparing two category amounts. A bar or column chart is suitable. A line connecting Paper to Writing can imply an order or progression that these labels do not have.

Follow with me once

On ChartSource, inspect A1:B3. The headers are category and amount; the two rows contain Paper 124.00 and Writing 90.00. Do not include a Grand Total as a third category: it already contains the first two and would double-count the meaning of the display.

Select A1:B3. Use Insert and choose a two-dimensional clustered bar or column chart. Give it the title "Order amount by category - 4 sample orders". Label the value axis "Amount (course currency units)" if the interface allows, or put that unit in a subtitle/caption beside the chart. These exercises use fictional amounts in one common currency; do not invent a real business currency that the source never stated.

Set the amount axis minimum to zero for this bar comparison. Bar length visually represents magnitude, so truncating its scale can exaggerate the difference. Remove three-dimensional effects, extra series and unnecessary decoration. Keep the category labels readable at normal zoom. Add data labels 124 and 90 if they help; do not force the reader to estimate every value from the scale.

Explain the chart in two sentences

"Paper has 124.00 of order amount and Writing has 90.00 in these four sample orders. Paper is higher by 34.00; this small delivery does not establish which category will lead next month." The first sentence reports observed values. The second gives a useful comparison and its scope. We have no costs, profit figures, customer preferences or future orders, so the chart does not establish profitability or future demand.

Compare the chart values with the source summary: $124 + 90 = 214$. A attractive picture can still come from the wrong range. Chart checking begins with the same source and reconciliation habits as a table.

Choose a different chart when the question changes

If the question is "How did daily order amounts change across a month?", a line chart with dates in order may help. If you only have two days, show the limited evidence honestly. For the current two-category comparison, two bars are enough. A pie chart is not required just because amounts can be expressed as shares.

If the chart looks wrong

If the category names appear as series with dates or IDs on the axis, inspect Select Data and the selected range. If a total bar dwarfs the groups, remove the grand total from the chart source. If labels overlap, widen the chart or use horizontal bars; do not shrink all text until it is unreadable on a phone. If values change, confirm whether the chart source or pivot needs refreshing.

Try it yourself

Create a two-column source with Paper 100, Writing 50 and Bags 250. Build a labelled bar or column chart without a grand-total category. State the highest category, the difference between Bags and Paper, the full total and one conclusion the chart cannot justify.

Changed retry - after your first review

The owner now asks for record counts: Paper 2, Writing 1, Bags 1. Create a second chart with an explicit record-count label. Explain why reusing the first chart's currency label would be wrong even though the category names match.

Revision in your own words

State the question, choose a suitable chart, label the measure and unit, check the source and respect the evidence's scope. For bars, start at zero. Exclude grand totals from category bars. Explain observed differences without inventing causes.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-13 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-14

Reconciliation, sanity checks and source preservation

Watch first

[1. Data Accuracy and Validation: Methods to ensure quality](#)

00:00-10:29

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-13. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Data Accuracy and Validation: Methods to ensure quality, 00:00-10:29.](#) [Optional](#) validation refresher. The lesson's row counts, total reconciliation and preserved-source handover go further than this general visual. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

The job of this lesson

Check a completed result from several useful directions and explain what remains uncertain. Complete DE01-13. Reconciliation means comparing quantities that should agree under your stated rules. A sanity check is a quick plausibility test; it can reveal obvious trouble but is not proof of correctness.

Words before we use them

Word	Plain meaning	Example
Control total	A total retained for comparison	original delivery amount 214.00

Difference One comparable quantity minus another
category sum minus full total

Reconciliation A comparison that should balance
group counts add to source count

Lineage A record of where a result came from

Raw to Orders to category summary

Exception A record needing attention missing paid status

Two formulas can agree because they share the same mistake. If both totals read the same incomplete range, their difference is zero even though the last order is missing. Whenever possible, compare with an independently established source count or a manually checked small

example, not just another reference to the same result cell.

Follow with me once

Return to the original four-order workbook, with all filters cleared. Keep Raw unchanged. In Orders, verify the four IDs against Raw. Inspect the formulas for 0007 and 0010, including the last row. Their amounts should be 99 and 60. Check each source quantity and price independently with a calculator if necessary.

Build a small check table beside the results:

Check	Expected	Observed for the original source
Source records preserved	4	4
Nonempty order IDs	4	4
Repeated order IDs	0	0
Whole amount	214.00	214.00
Paid + unpaid + unknown	whole amount	124 + 30 + 60 = 214
Payment group counts	4	2 + 1 + 1 = 4
Unknown status records	reported separately	1, order 0010

Write the check and the source of its expected value. "Expected 214 because another total cell says 214" is weak when both cells may share an omitted row. Here the four manual row calculations provide a small independent basis.

Break one thing on a copy

Copy the completed workbook as reconciliation-error.xlsx. Change its unknown payment group to Unpaid without changing the raw status. Whole amount remains 214. Paid plus unpaid plus unknown still adds to 214. Yet the source-to-category comparison fails: order 0010 is missing in the source but called Unpaid in the result. Record this deliberate error, then repair the formula from the actual status and retest.

Now imagine a different fault: order 0010 was omitted entirely. The remaining total would be 154.00 and record count 3. Category subtotals could still add perfectly to 154.00. Comparing against the original four-row delivery catches what an internal category-sum check alone can miss.

Preserve an inspectable handover

Keep the source file, your working XLSX, the rules used and a short result explanation together. Name the source and its sample dates. Record unresolved unknowns instead of burying them in a total. Use a different filename for a changed attempt; an engineer should be able to trace which inputs produced a submitted result.

If checks disagree

Start with the smallest failed comparison. A count difference suggests missing, extra or filtered records; an amount difference with matching counts suggests values, types, formulas or grain; a category mismatch with matching total suggests classification. These are investigation leads, not automatic diagnoses. Inspect affected rows before changing values.

Try it yourself

For Independent, reconcile source count 4, row amounts 60/50/40/250, full total, payment totals and payment counts. State which check would fail if the unknown order 0013 were dropped, and which would fail if it were kept but misclassified as Unpaid.

Changed retry - after your first review

A result keeps four rows and total 400, but swaps the recorded paid statuses of orders 0011 and 0012. Explain whether record count and whole total detect it, and calculate the new paid and unpaid amounts to show the consequence.

Revision in your own words

Compare with a preserved source and state each check's purpose. Check counts, keys, values, formulas and categories. A zero difference supports one comparison. It does not erase unknowns or prove that the real-world source is accurate.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-14 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.

DE01-15

A complete small workflow: inspect, calculate, validate, explain and save

Watch first

[1. Data Journey: What you need to know for successful navigation](#)

00:00-04:37

Pause to connect the demonstration to the example below. Use the printed stop time when a bounded range is given; longer recordings remain available for reference.

Before starting: DE01-14. Video: Instructor watch ranges appear below. Read the vocabulary, perform the guided example and save an independent attempt before review.

[Instructor demonstration: Data Journey: What you need to know for successful navigation, 00:00-04:37](#). Picture the data workflow from beginning to end, then build the book's independent small handover. Stop manually at the listed end. Original educational video page, published full runtime and accompanying transcript inspected; full playback not verified. Open the page and play its embedded video from the beginning.

The job of this lesson

Carry a fresh small delivery from source to an inspectable handover. Complete DE01-14. This is your first integrated workflow. Do it slowly enough to explain each stage. Speed is not the assessment. Use the MiniWorkflow sheet in a new copy of the practice workbook and keep that sheet unchanged as your input.

The owner's request

"For these orders, tell me the whole order amount, the amount recorded paid, the amount recorded unpaid and the amount whose status is unknown. Show enough checks that another person can inspect your work." One row means one order containing one product kind. Unit price is for one item. All amounts share a currency. Ignore taxes, discounts and delivery. Unknown status is permitted and must remain separate.

order_id	item	quantity	unit_price	ordered_on	is_paid
B001	Notebook	3	40.00	2026-09-08	TRUE
B002	Pen	5	8.00	2026-09-08	FALSE

order_id	item	quantity	unit_price	ordered_on	is_paid
B003	Folder	2	25.00	2026-09-09	(missing)
B004	Marker	1	30.00	2026-09-09	TRUE

Follow the sequence, with reasons

First inspect. Confirm four source records, six fields, nonempty unique text IDs, positive whole quantities, numeric prices and the allowed payment answers. The source contains one unknown status. Record that observation before calculations; it is part of the result, not a nuisance to clean away.

Second create a working copy called MiniWork. Add amount and payment_group columns using the formulas learned earlier. Calculate one row manually before filling down: B001 is $3 \times 40 = 120$. Inspect the last copied formula as well as the first. The later row amounts are 40, 50 and 30, giving a whole total of 240.

Third summarise by the requested payment groups. Paid combines B001 and B004, so $120 + 30 = 150$. Unpaid is B002 at 40. Unknown is B003 at 50. Counts are 2, 1 and 1. Keep the classification tied to each order's original status. A blank does not become FALSE.

Fourth validate. Compare the working IDs and row count with MiniWorkflow. Reconcile $150 + 40 + 50$ to 240 and $2 + 1 + 1$ to 4. Inspect B003 specifically to confirm it remains unknown. Verify that you multiplied quantity by price rather than summing prices. Describe what these checks establish: consistency with this supplied delivery and the stated rules, not independent proof that every payment record matches reality.

Fifth explain and save. Write: "Across four supplied sample orders dated 8-9 September 2026, total order amount is 240.00. Recorded paid is 150.00, recorded unpaid is 40.00 and unknown status is 50.00. Counts and amounts reconcile; B003 still needs payment confirmation." Use your own wording if it preserves these facts. Save as DE01-15-guided.xlsx under evidence, close and reopen it. Save the file path, checks and explanation in your attempt record. Keep a backup of both the record and workbook; writing the path does not copy the workbook.

What makes the handover usable

Another reader needs the source, the working result, the interpretation rules, the checks and the unresolved issue. A screenshot of a grand total supplies only one small piece. The workbook's formulas help them inspect the calculation, while the written explanation tells them what question it answers.

If you cannot finish a stage

Repair that stage before moving on. Wrong row meaning: return to DE01-02. Lost leading zeros: DE01-06. Broken copied formulas: DE01-08 or 09. Wrong unknown classification: DE01-03 and 08. Mismatched totals: DE01-14. Your revision notes already contain these explanations; you do not need to search for another course.

Try it yourself

Use the FinalAttempt sheet: C001 Notebook 2 at 55 TRUE; C002 Pen 6 at 7.50 FALSE; C003 Bag 1 at 300 missing; C004 Ruler 3 at 15 TRUE, all dated 2026-09-10. Complete the entire workflow in a copy. Submit the workbook, row calculations, groups, counts, reconciliation, an issue note and a short owner-facing explanation. Save an attempt before opening Review.

Changed retry - after your first review

Use FinalRetry: D001 Pencil 4 at 9 TRUE; D002 Eraser 3 at 6 FALSE; D003 Folder 2 at 22 missing, all dated 2026-09-11. The new question asks for the amount with a known payment status, including both TRUE and FALSE, plus unknown separately. Complete a new handover and explain how the inclusion rule changed.

Revision in your own words

Inspect the source and grain; preserve it; calculate with references; classify by explicit rules; validate independently; explain scope and unknowns; save and reopen the evidence. Passing this module shows beginner data-handling skills. It does not yet establish SQL, pipeline or job readiness. The next stage builds on this foundation.

Evidence to keep: your attempt, the observed result and its explanation. Save the first attempt before opening DE01-15 in the Explained Reviews book. After review, solve the changed retry and keep it as a separate attempt.